Name: B.Vinayani
Registration Number: 192472179

EXPERIMENT 12

Prompt Evaluation, Comparison and Refinement using LLM Responses

AIM

To design multiple prompts for a text-generation task, evaluate their outputs using predefined criteria (relevance, accuracy, completeness, clarity and format adherence) and refine the best-performing prompt.

PROCEDURE

1. Select the task: write a short note on the benefits of regular exercise for first-year college students.
2. Design three prompts of increasing detail: P1 (basic), P2 (adds role and audience) and P3 (adds word limit and exact structure).
3. Define the evaluation criteria and a 1-5 scoring scale: relevance, accuracy, completeness, clarity and format adherence (maximum 25).
4. Generate the output of each prompt and score it against the criteria (the program uses the LLM as a judge and also counts words and bullet points automatically).
5. Select the prompt with the highest total and note its weakest criteria.
6. Refine the best prompt to fix the weak criteria (P4), generate its output, score it again and compare it with the original.

PROMPTS AND CODE

Prompt P1 (basic):

```
Write about the benefits of exercise for students.
```

Prompt P2 (role and audience):

```
You are a college health counsellor. Write a short, friendly note for first-year
college students explaining the benefits of regular exercise.
```

Prompt P3 (role, audience and format):

```
You are a college health counsellor writing for first-year students. Write a note
of at most 100 words with: a one-line opening, exactly three bullet points
(physical, mental and academic benefits) and a one-line closing call to action.
Use simple, friendly language and avoid medical jargon.
```

Prompt P4 (refined version of P3):

```
You are a college health counsellor writing for first-year students. Write a note
of at most 100 words with: a one-line opening, exactly three bullet points
(physical, mental and academic benefits) and a one-line closing call to action.
Use simple, friendly language and avoid medical jargon. Keep each bullet under 20
```

```
words, stay strictly within 100 words, and end with one practical tip, for
example a 20-minute daily routine.
```

exp12_prompt_eval.py

```
"""Experiment 12 - Prompt evaluation, comparison and refinement.

Several prompts for the same text-generation task are run, scored against a
rubric (relevance, accuracy, completeness, clarity, format adherence) and the
best prompt is refined using the weaknesses found in its evaluation.
"""
import json
import os
import re
import sys

from openai import OpenAI

MODEL = "gpt-4o-mini"
CRITERIA = ["relevance", "accuracy", "completeness", "clarity",
"format_adherence"]

PROMPTS = {
    "P1": "Write about the benefits of exercise for students.",
    "P2": ("You are a college health counsellor. Write a short, friendly note for
first-year "
           "college students explaining the benefits of regular exercise."),
    "P3": ("You are a college health counsellor writing for first-year students.
Write a note of "
           "at most 100 words with: a one-line opening, exactly three bullet
points (physical, "
           "mental and academic benefits) and a one-line closing call to action.
Use simple, "
           "friendly language and avoid medical jargon."),
}

RUBRIC = """You are a strict evaluator. Score the TEXT from 1 (poor) to 5
(excellent) on each criterion:
- relevance: stays on the topic and suits first-year college students
- accuracy: every statement is factually correct
- completeness: covers physical, mental and academic benefits and ends with a
call to action
- clarity: simple, friendly and easy to read
- format_adherence: at most 100 words, one-line opening, exactly three bullet
points, one-line closing
Return only JSON with the keys relevance, accuracy, completeness, clarity,
format_adherence."""


def chat(client, content, json_mode=False, temperature=0.7):
    kwargs = {"response_format": {"type": "json_object"}} if json_mode else {}
    reply = client.chat.completions.create(
        model=MODEL, temperature=temperature,
        messages=[{"role": "user", "content": content}], **kwargs)
    return reply.choices[0].message.content.strip()


def auto_checks(text):
    """Objective format checks: word count and number of bullet points."""
    words = len(text.split())
    bullets = len(re.findall(r"^\s*[-*•]", text, re.MULTILINE))
    return words, bullets


def evaluate(client, name, prompt):
    text = chat(client, prompt)
```

```
    scores = json.loads(chat(client, f"{RUBRIC}\n\nTEXT:\n{text}",
json_mode=True, temperature=0))
    words, bullets = auto_checks(text)
    return {"name": name, "prompt": prompt, "text": text, "scores": scores,
            "total": sum(scores[c] for c in CRITERIA), "words": words, "bullets":
bullets}


def print_table(rows):
    print(f"{'Prompt':<8}{'Words':>6}{'Bullets':>8}{'Rel':>5}{'Acc':>5}{'Cmp':>5}
{'Clr':>5}{'Fmt':>5}{'Total':>9}")
    for r in rows:
        s = r["scores"]
        print(f"{r['name']:<8}{r['words']:>6}{r['bullets']:>8}{s['relevance']:>5}
{s['accuracy']:>5}"
              f"{s['completeness']:>5}{s['clarity']:>5}{s['format_adherence']:>5}
{str(r['total']) + '/25':>9}")


def refine(client, best):
    weak = sorted(CRITERIA, key=lambda c: best["scores"][c])[:2]
    request = (
        "Improve the PROMPT below so that its output scores higher on these weak
criteria: "
        f"{', '.join(weak)}. The output currently has {best['words']} words and
{best['bullets']} bullets. "
        "Keep the task and the audience the same. Return only the improved
prompt.\n\n"
        f"PROMPT:\n{best['prompt']}")
    return chat(client, request, temperature=0.3), weak


def main():
    key = os.getenv("OPENAI_API_KEY")
    if not key:
        sys.exit("Error: set the OPENAI_API_KEY environment variable first.")
    client = OpenAI(api_key=key)

    print("=== Round 1: evaluating prompts ===")
    rows = [evaluate(client, name, prompt) for name, prompt in PROMPTS.items()]
    print_table(rows)
    best = max(rows, key=lambda r: r["total"])
    print(f"\nBest prompt: {best['name']} ({best['total']}/25)")

    refined_prompt, weak = refine(client, best)
    print(f"Weakest criteria: {', '.join(weak)}")
    print("\n=== Refined prompt (P4) ===")
    print(refined_prompt)

    print("\n=== Round 2: evaluating refined prompt ===")
    refined = evaluate(client, "P4", refined_prompt)
    print_table([best, refined])
    print(f"\nImprovement: {best['name']} {best['total']}/25 -> P4
{refined['total']}/25")


if __name__ == "__main__":
    main()
```

OUTPUT

Exp 12 - Prompt P1: Basic prompt (sample output)

PROMPT

Write about the benefits of exercise for students.

MODEL OUTPUT

Exercise is very important for students. It helps keep the body healthy and strong, and it can reduce the risk of many diseases in the future. Regular physical activity also improves mood by releasing chemicals in the brain that make people feel happier and less stressed. Students who exercise often sleep better, which helps them stay alert in class and remember what they study. Exercise can also build discipline and time management, because students must plan their day to include a workout. Joining a sports team or a gym can help students make new friends and feel more connected to college life. In addition, regular activity boosts energy levels, so students feel less tired during long study sessions. Overall, exercise is good for both the body and the mind.

Figure: Prompt P1 (basic prompt) and sample output

Exp 12 - Prompt P2: Role and audience (sample output)

PROMPT

You are a college health counsellor. Write a short, friendly note for first-year college students explaining the benefits of regular exercise.

MODEL OUTPUT

Hello and welcome to college! As your health counsellor, I would love to share why regular exercise deserves a place in your timetable. Moving your body strengthens your heart and muscles and gives you more energy for busy days. It also lifts your mood and eases the stress that comes with deadlines and exams. Many students notice that they sleep better and concentrate more when they stay active. Exercise is also a wonderful way to meet new people, whether you join a sports club or simply walk with a friend. You do not need a gym or special equipment; even a short daily walk helps. Start small, stay consistent, and your body and mind will thank you.

Figure: Prompt P2 (role and audience) and sample output

Exp 12 - Prompt P3: Role, audience and format (sample output)

PROMPT

You are a college health counsellor writing for first-year students. Write a note of at most 100 words with: a one-line opening, exactly three bullet points (physical, mental and academic benefits) and a one-line closing call to action. Use simple, friendly language and avoid medical jargon.

MODEL OUTPUT

Welcome to college, and congratulations on starting this exciting new chapter of your life!
- Physical: Regular exercise strengthens your heart, muscles and bones, and gives you more energy for long days of classes, labs and activities.
- Mental: Staying active lifts your mood, reduces stress and helps you feel calmer before exams and tight deadlines, so you can enjoy campus life more.
- Academic: Exercise improves sleep, focus and memory, which makes studying more effective and helps you learn faster and perform better in tests.

Make movement a part of your day, because a healthier you is a better student, a better friend and a happier person!

Figure: Prompt P3 (role, audience and format) and sample output

Exp 12 - Prompt P4: Refined prompt (sample output)

PROMPT

You are a college health counsellor writing for first-year students. Write a note of at most 100 words with: a one-line opening, exactly three bullet points (physical, mental and academic benefits) and a one-line closing call to action. Use simple, friendly language and avoid medical jargon. Keep each bullet under 20 words, stay strictly within 100 words, and end with one practical tip, for example a 20-minute daily routine.

MODEL OUTPUT

Welcome to college! Here is a simple habit that pays off all year.
- Physical: Exercise strengthens your heart and muscles and boosts your daily energy.
- Mental: Moving your body lifts your mood and eases exam stress.
- Academic: Better sleep and sharper focus make studying more effective.

Tip: Start with a 20-minute walk or workout each day, and let your body and mind thank you!

Figure: Prompt P4 (refined prompt) and sample output

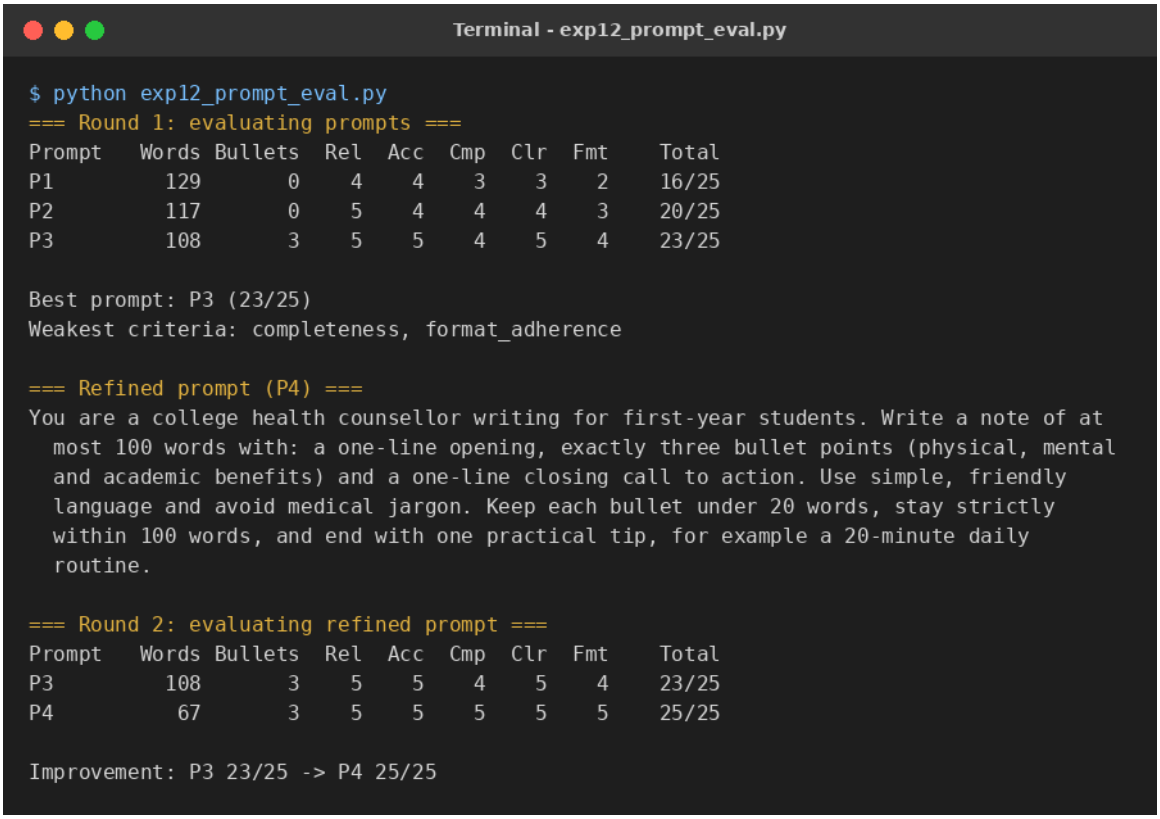


Figure (program) 1: Sample output of the program

Evaluation of the prompts (score 1-5 for each criterion)

Criteria	P1 Basic	P2 Role + audience	P3 Role + format	P4 Refined
Relevance	4	5	5	5
Accuracy	4	4	5	5
Completeness	3	4	4	5
Clarity	3	4	5	5
Format adherence	2	3	4	5
Total (out of 25)	16	20	23	25
Words / bullets	129 / 0	117 / 0	108 / 3	67 / 3
Main observation	Generic, long, no structure	Friendly tone, but one paragraph and no fixed format	Right structure, but 108 words (over the limit) and no practical tip	Within limit, 3 bullets and a practical tip

RESULT

P1 scored 16/25, P2 scored 20/25 and P3 scored 23/25, so P3 was the best prompt. Its weaknesses were completeness (no practical tip) and format adherence (108 words against a limit of 100). The refined prompt P4 added a per-bullet length limit, a strict word limit and a practical-tip requirement; its output had 67 words, three bullets and a tip, and scored 25/25.

CONCLUSION

Prompt quality can be measured by scoring outputs against fixed criteria. Adding role, audience, structure and limits steadily improved the responses, and refining the best prompt using the weakest criteria gave a further improvement. Evaluating and refining prompts in this way is a systematic route to reliable LLM output.